

Predictive Analysis of Suicide Rates in the European Union

Socioeconomic and mental-health determinants, panel modelling and temporal persistence — EU-27, 2000-2023

Data sources: WHO • World Bank • IHME

Select region

All



Select year

All



27

Countries

2000-2021

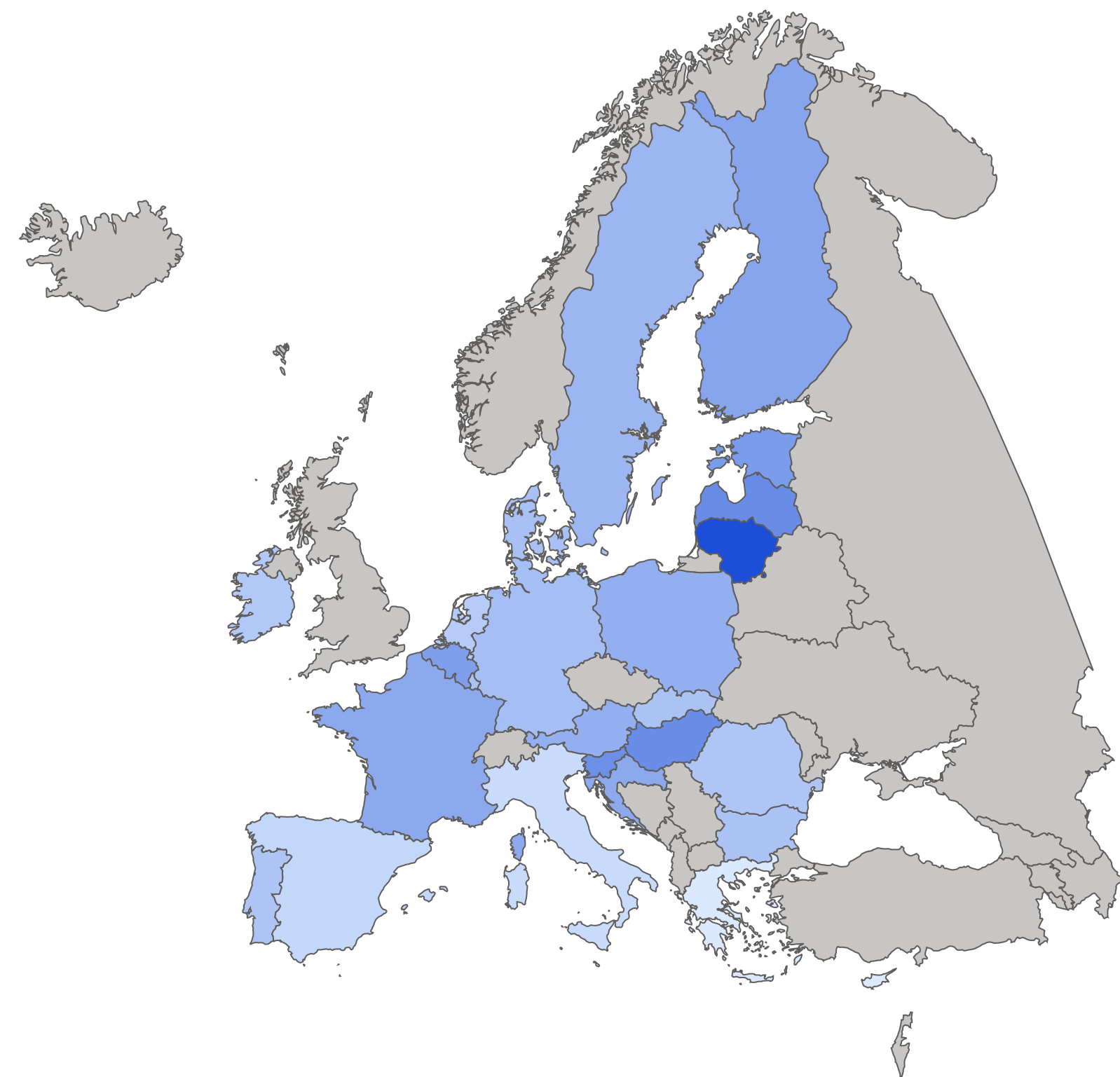
Years covered

14.86

Avg suicide rate
(per 100k)

CatBoost

Selected model



Key insights

- Alcohol use is the strongest predictor of suicide rates.
- Baltic countries exhibit the highest suicide rates across the EU.
- Splitting by time generalises far better than splitting by country.
- CatBoost selected as the production model due to its overall robustness.
- Mental health provides more predictive power than socioeconomic variables.
- Suicide rate is highly self-persistent - only one exogenous determinant added to SARIMAX meaningfully beats the naive baseline.

Executive Summary

Suicide landscape

Drivers of suicide rates

Model performance

Clustering analysis

Time forecasting models

Conclusions

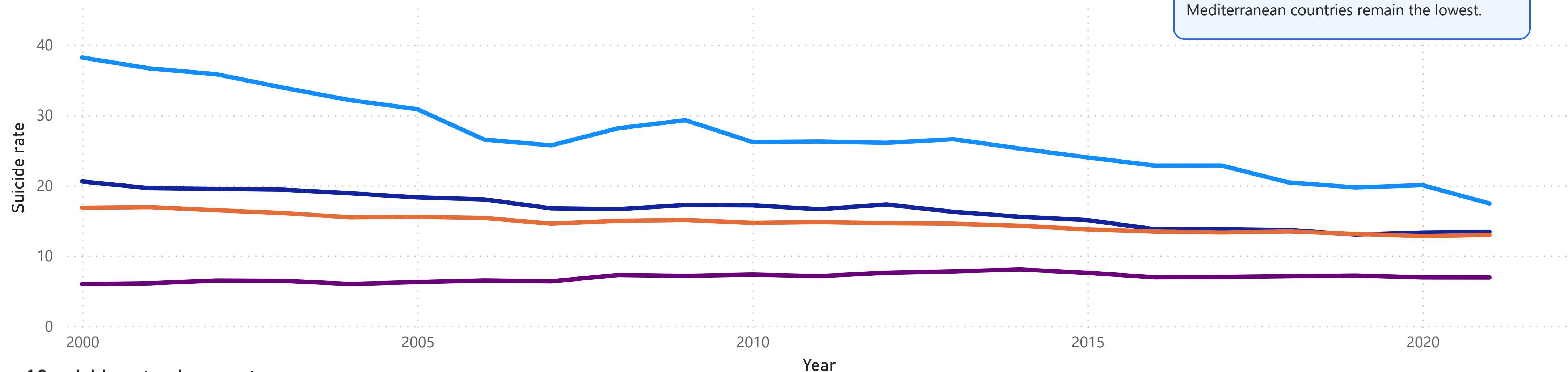
Suicide landscape in Europe

Choose country

All

Suicide rate evolution

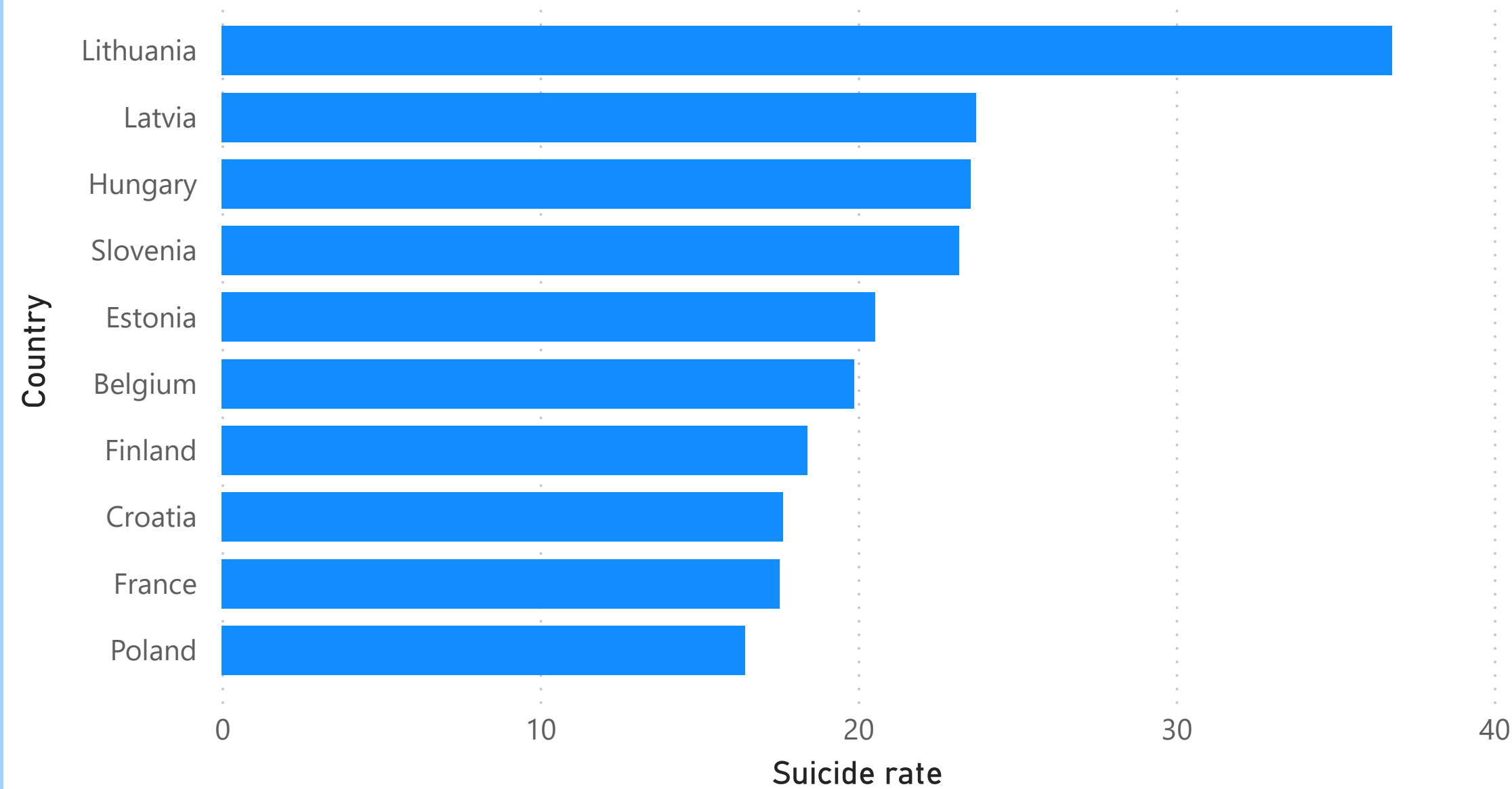
Baltics Eastern Europe Mediterranean Western Europe/Nordics



Key insights

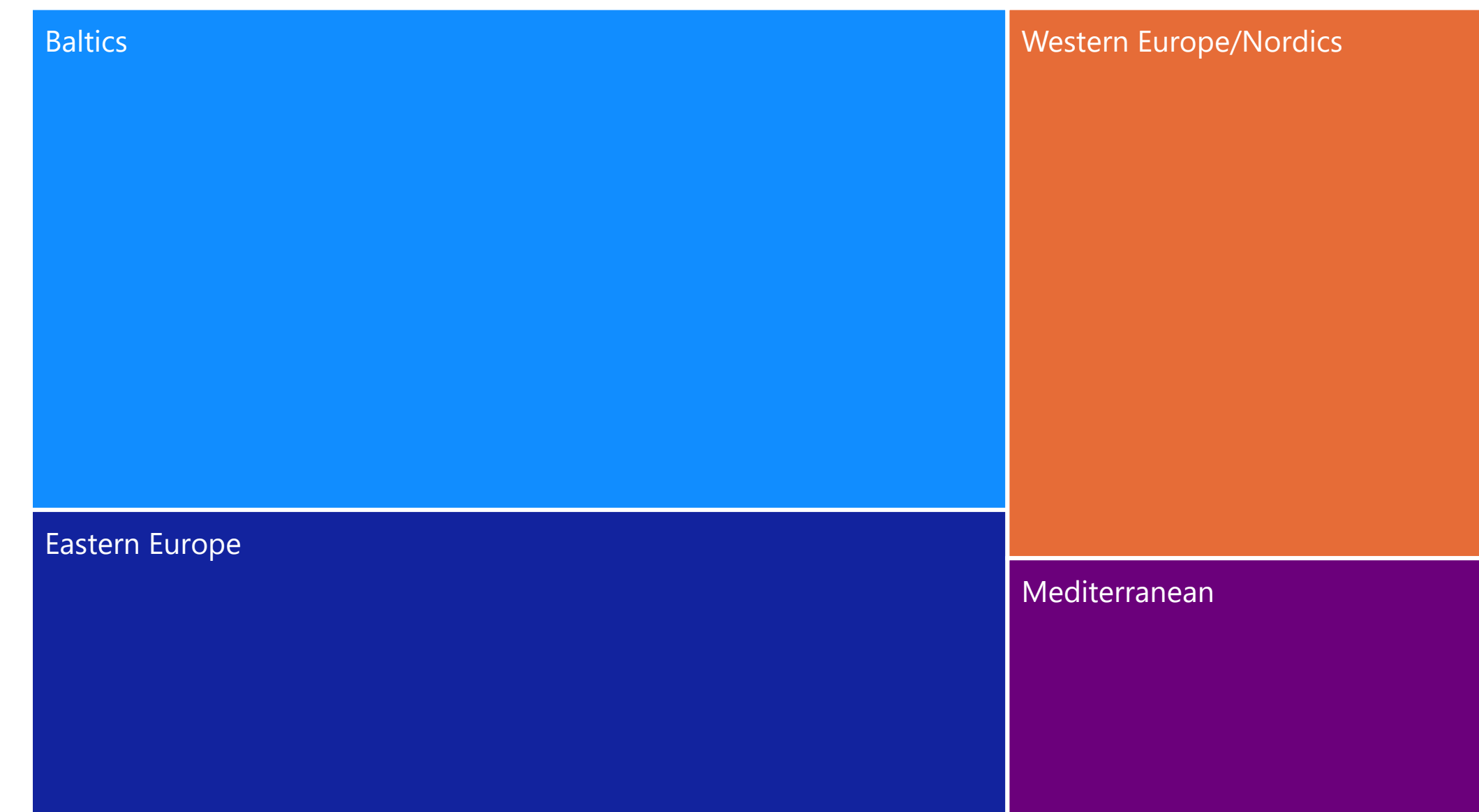
Baltic countries consistently show the highest suicide rates across the study period, while Mediterranean countries remain the lowest.

Top 10 suicide rates by country



Year

Average Suicide Rate by Region (2000-2021)



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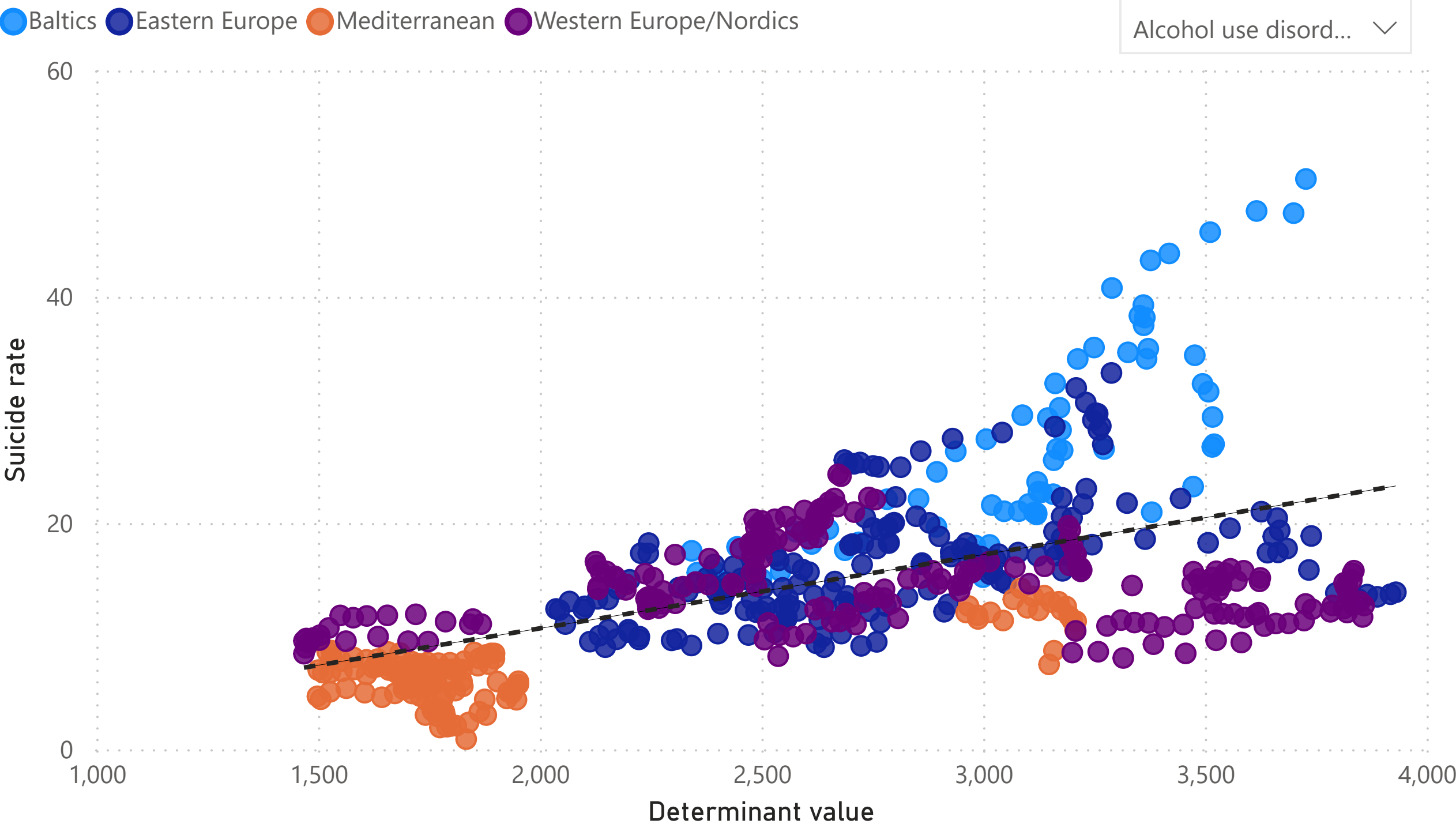
Drivers of suicide rates

Select region

All



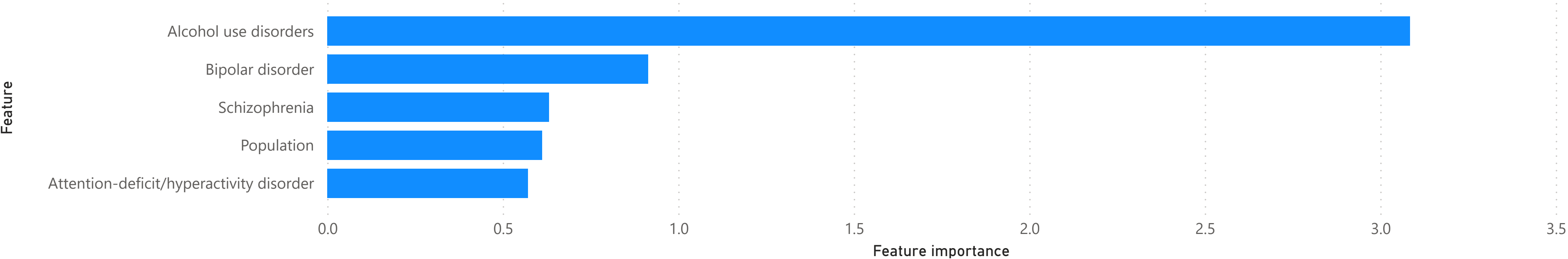
Relationship between determinants and suicide rate



Key insights

- Alcohol Use Disorders emerge as the strongest predictor of suicide rates.
- Mental health variables consistently outperform classical socioeconomic indicators.
- Bipolar Disorder and Schizophrenia show significant predictive contribution.
- GDP and unemployment exhibit substantially lower predictive power than expected.
- Results support the hypothesis that mental-health prevalence provides stronger explanatory value than macroeconomic conditions.

SHAP feature importance



Model performance

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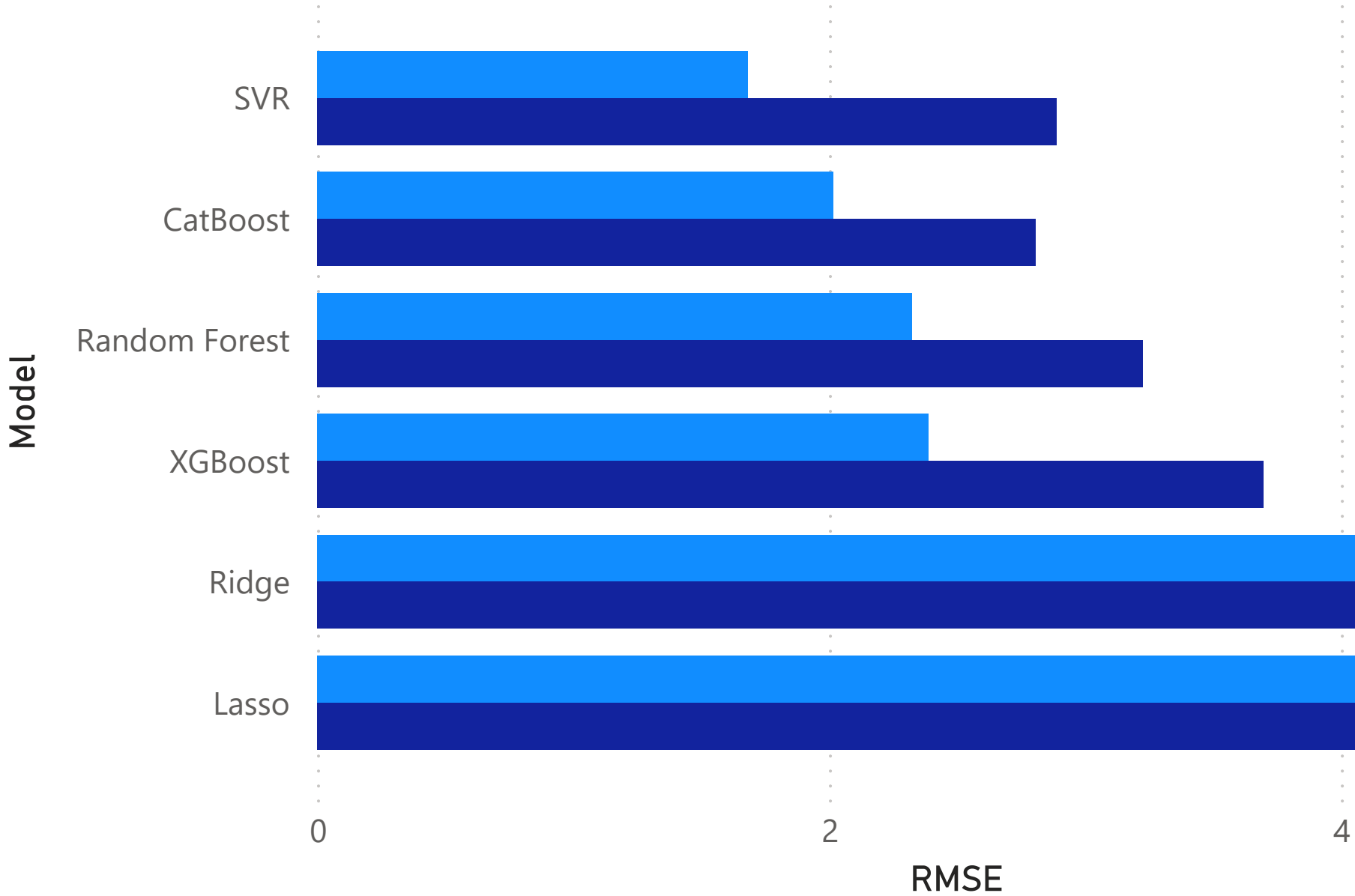
Time forecasting models

Conclusions

Model	Split	CV RMSE	MAE	RMSE	R ²	Time (s)
CatBoost	Test	2.14	1.47	2.02	0.88	26.41
CatBoost	Validation	2.14	2.19	2.81	0.64	26.41
Lasso	Test	5.09	3.89	4.70	0.33	0.05
Lasso	Validation	5.09	4.06	4.78	-0.03	0.05
Random Forest	Test	2.54	1.72	2.32	0.84	21.65
Random Forest	Validation	2.54	2.53	3.22	0.53	21.65
Ridge	Test	5.11	3.82	4.62	0.36	0.05
Ridge	Validation	5.11	4.20	4.88	-0.08	0.05
SVR	Test	1.75	1.33	1.68	0.91	0.91
SVR	Validation	1.75	2.21	2.89	0.62	0.91
XGBoost	Test	2.60	1.69	2.39	0.83	3.42
XGBoost	Validation	2.60	2.69	3.70	0.38	3.42

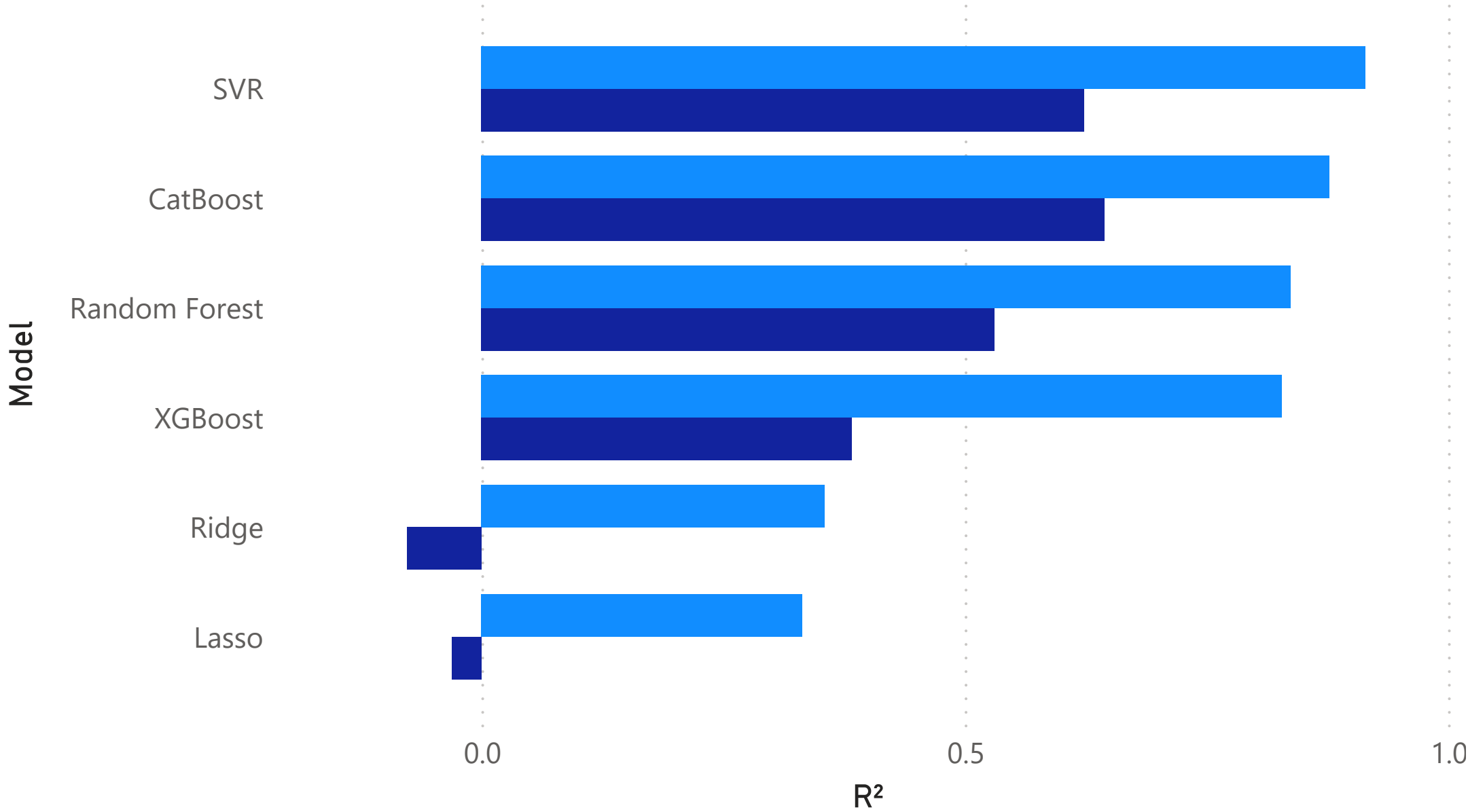
RMSE by model

● Test ● Validation



R² by model

● Test ● Validation



CatBoost

Selected model

0.64

Validation R²

2.81

Validation RMSE

Key insights

- CatBoost and SVR achieve the strongest predictive performance.
- Linear models (Ridge and Lasso) show limited predictive capacity.
- CatBoost was selected as the production model due to its strong validation performance, robustness and deployment suitability.
- Results demonstrate that suicide rates can be predicted with meaningful accuracy using health and socioeconomic indicators.

Select model

All

Select option

Option B

Select split

All

Clustering analysis

k=4

Clusters selected

0.57

ARI score

0.72

NMI score

Select region

All



Select cluster

All



Key insights

- K-Means and Hierarchical Clustering partially validate the predefined regional grouping.
- Baltic and Eastern European countries form the most clearly defined clusters.
- Mediterranean countries overlap with parts of Western Europe and Nordic countries.
- Economic scale emerges as a stronger clustering factor than geography alone.
- Clustering confirms the existence of structural differences across European countries.

Executive Summary

Suicide landscape

Drivers of suicide rates

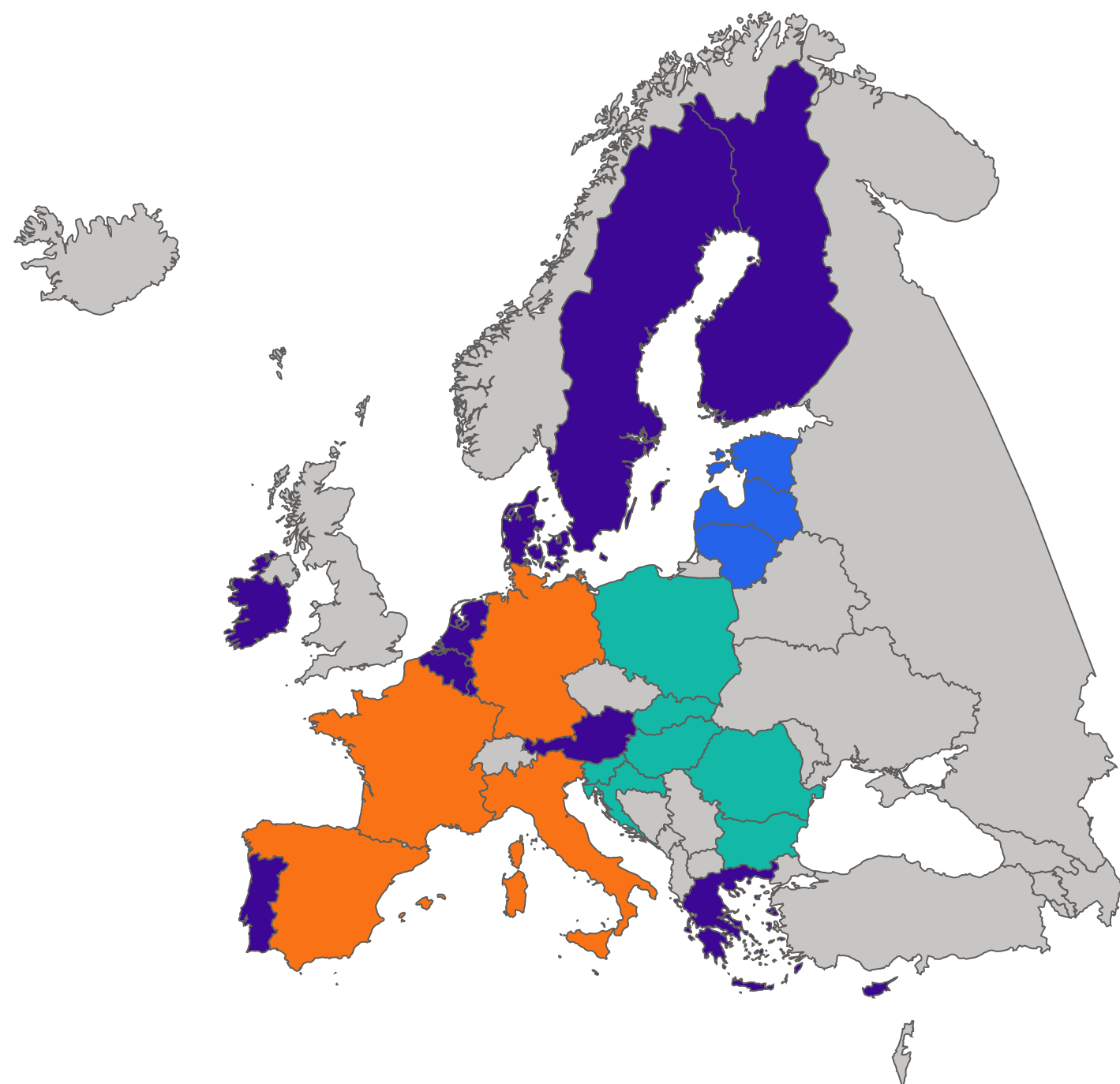
Model performance

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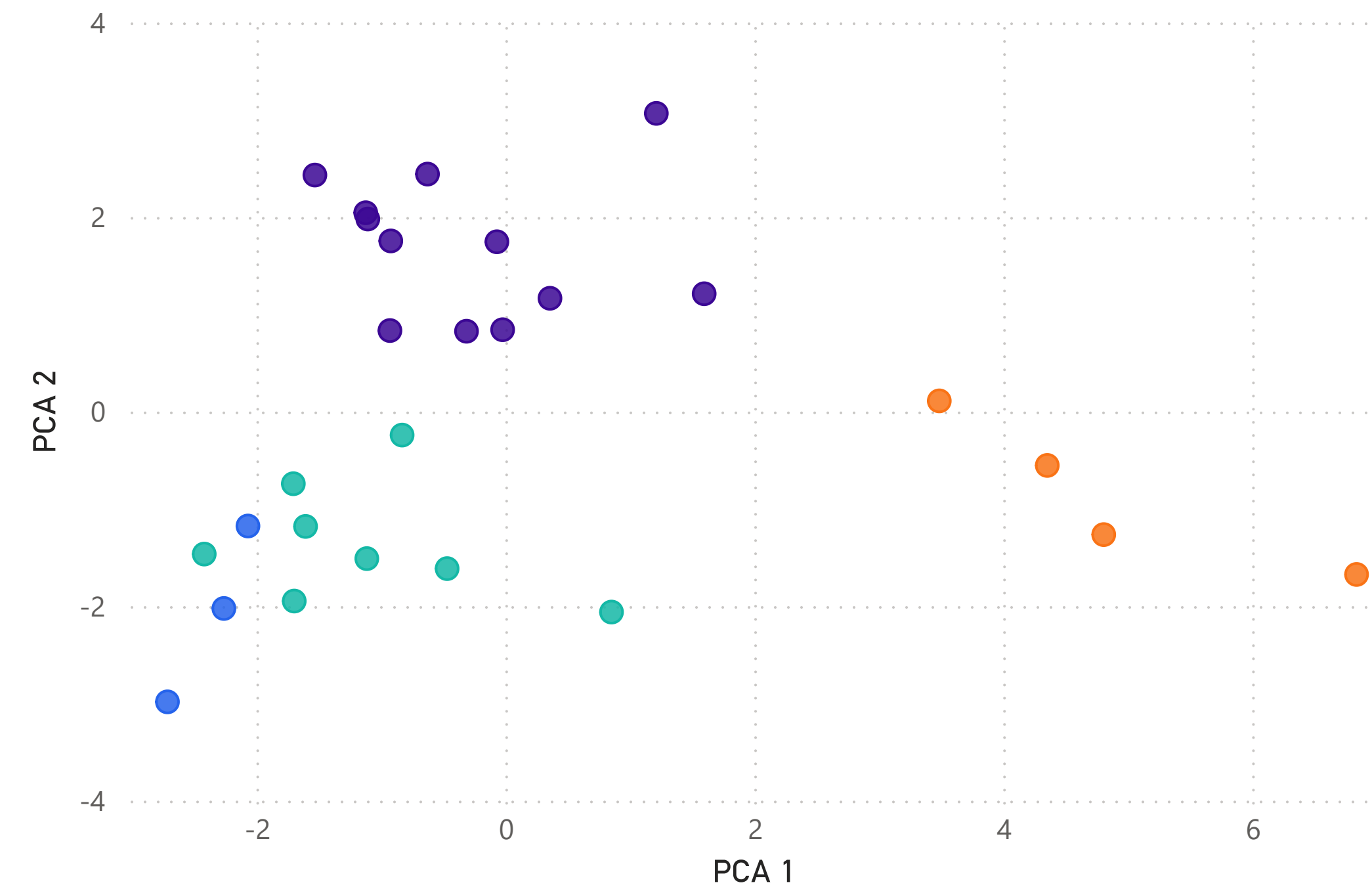
Conclusions

- Cluster 3
- Cluster 1
- Cluster 0
- Cluster 2



Countries in PCA cluster space

Cluster 0 Cluster 1 Cluster 2 Cluster 3



Cluster composition



Time forecasting models

SARIMAX + exog
Selected forecasting model

0.77
Validation R²

2.28
Validation RMSE

Select model

All

Select variant

All

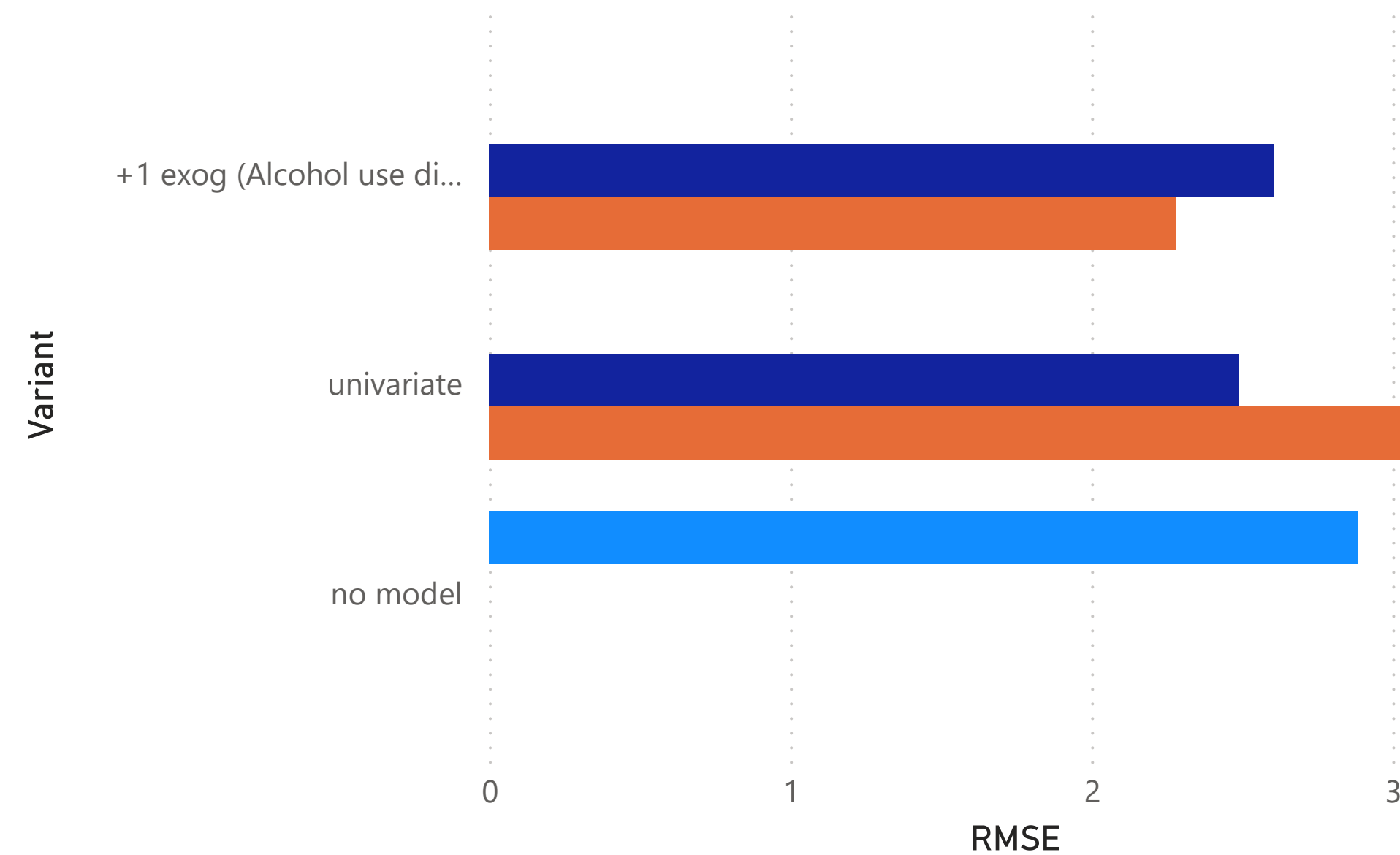
Model	Variant	Test RMSE	Val RMSE	Test R ²	Val R ²
SARIMAX	+1 exog (Alcohol use disorders)	1.41	2.28	0.94	0.77
Prophet	univariate	1.79	2.49	0.90	0.72
Prophet	+1 exog (Alcohol use disorders)	1.72	2.60	0.91	0.69
Naive persistence	no model		2.88		0.62
SARIMAX	univariate	1.78	3.20	0.90	0.54

Key insights

- Temporal persistence explains a substantial proportion of suicide-rate variability.
- Univariate SARIMAX significantly underperforms naive forecasting approaches.
- Introducing Alcohol Use Disorders as an exogenous variable improves predictive performance.
- Forecasting models confirm the long-term stability of regional suicide patterns.
- Time-series models provide a strong complement to supervised machine learning approaches.
- SARIMAX with Alcohol Use Disorders achieves the strongest validation performance.

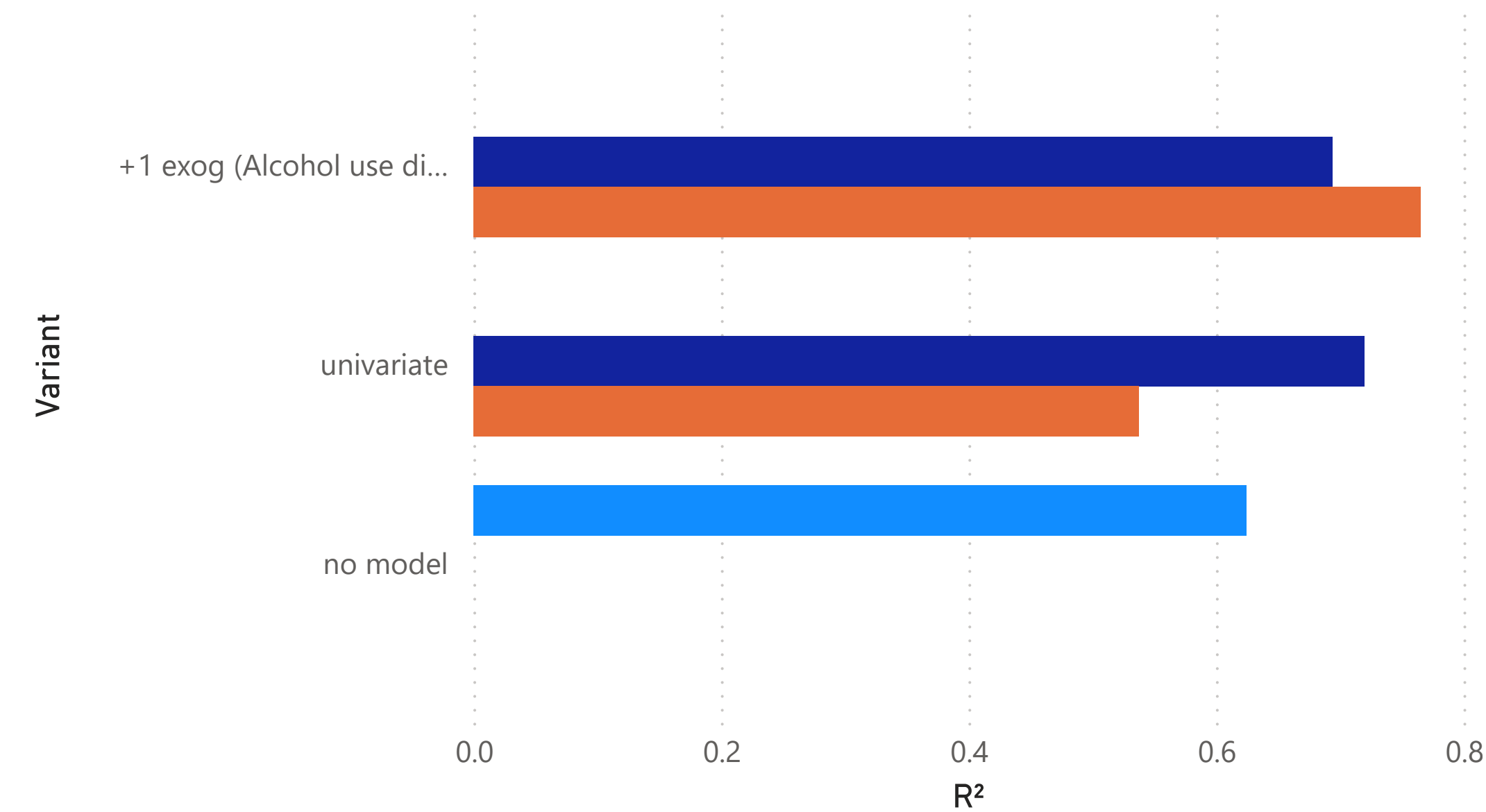
Validation RMSE by forecasting model

Naive persistence Prophet SARIMAX



Validation R² by forecasting model

Naive persistence Prophet SARIMAX



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Final conclusions

- Suicide rates in Europe are not randomly distributed and display strong regional patterns.
- Mental-health conditions are considerably more informative than classical socioeconomic indicators.
- Alcohol Use Disorders consistently emerged as the most important determinant of suicide rates.
- CatBoost provided the strongest supervised-learning performance, while SARIMAX achieved the best temporal forecasting results.
- The combination of machine learning, clustering and forecasting techniques successfully identified meaningful patterns and generated accurate predictions.
- These findings highlight the potential of data-driven approaches to support suicide prevention strategies and public-health decision making.

Overall Outcome

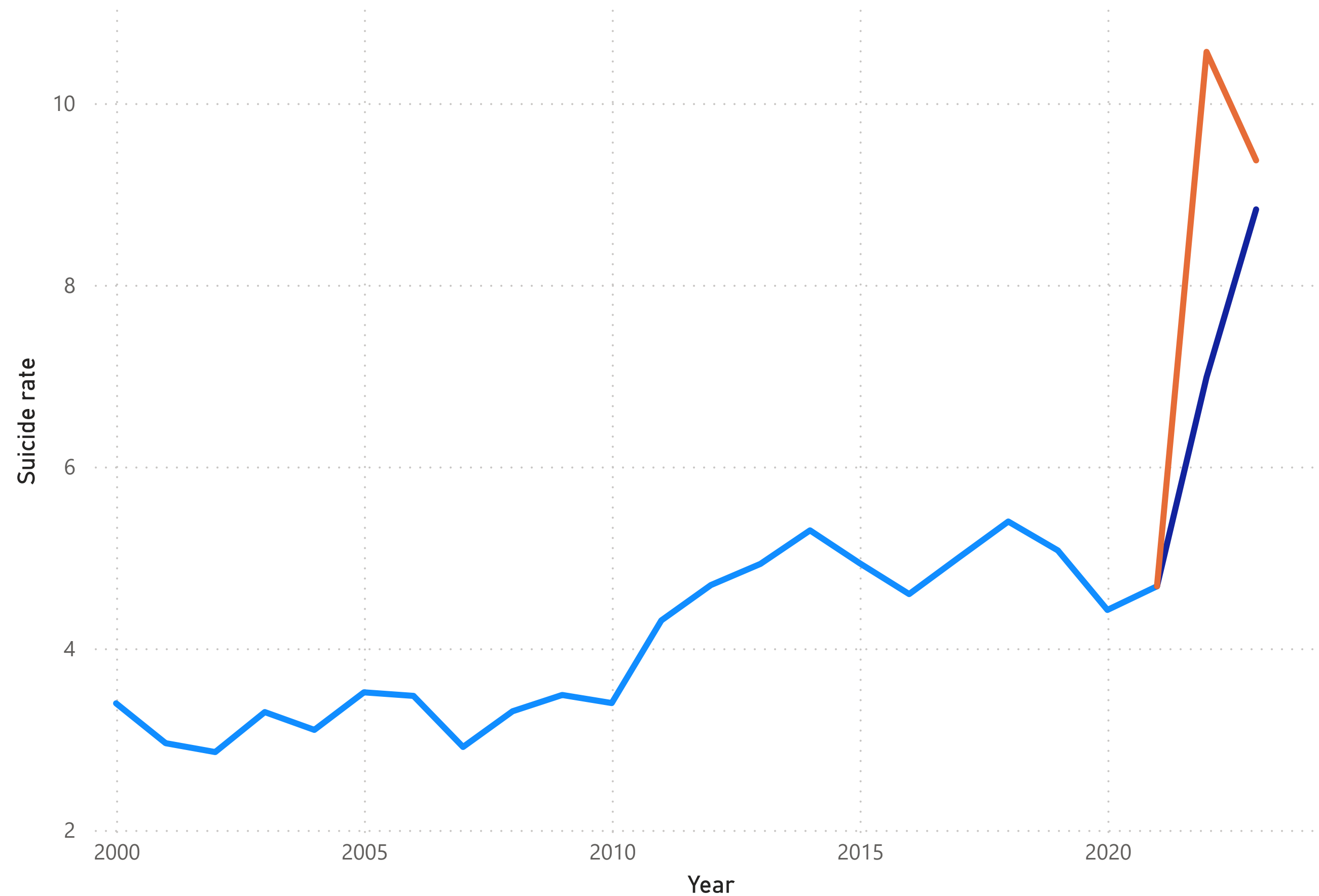
The combination of Machine Learning, Clustering and Time-Series Forecasting provides a robust framework for understanding and predicting suicide rates across Europe.

Historical trends and forecasts

● Actual ● CatBoost ● SARIMAX +1 exog

Select country

Greece



Alcohol use disorders

Strongest predictor

CatBoost

Strongest ML model

SARIMAX + exog

Strongest forecasting model